On my honor I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues.

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1)

a) $\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix}$

$k = 14 \quad \alpha = 0 \quad \gamma = 0$

$\alpha^2 = 0 \Rightarrow \alpha = 0$

$\beta = -1$

$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$

$k = 14 \quad \alpha = 0 \quad \gamma = 0 \quad \beta = -1 \quad m = 2$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 5k \\ k \\ 9k-6 \end{pmatrix} = \begin{pmatrix} 5k \\ k \\ 9k \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ -6 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = k\begin{pmatrix} 5 \\ 1 \\ 9 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ -6 \end{pmatrix} ; \quad k \in \mathbb{R}$$

=) vectors in their intersection:

$$\left[k\begin{pmatrix} 5 \\ 1 \\ 9 \end{pmatrix} : \begin{pmatrix} 0 \\ 0 \\ -6 \end{pmatrix} \right] : k \in \mathbb{R} \quad \text{for } k = 1$$

vectors: $\left\{ \begin{pmatrix} 5 \\ 1 \\ 9 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ -6 \end{pmatrix} \right\}$

Clearly $\begin{pmatrix} 5 \\ 1 \\ 9 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \\ -6 \end{pmatrix}$ are linearly independent as they satisfy $\alpha\begin{pmatrix} 5 \\ 1 \\ 9 \end{pmatrix} + \beta\begin{pmatrix} 0 \\ 0 \\ -6 \end{pmatrix}$

$\Downarrow$

$\textcircled{0}$

only for $\alpha = \beta = 0$

=> Vectors $\left\{ \begin{pmatrix} 5 \\ 1 \\ 9 \end{pmatrix} ; \begin{pmatrix} 0 \\ 0 \\ -6 \end{pmatrix} \right\}$ forms a vectors space

It has a dimension of 2 inside 3 dimensional space of proposed geometry.

Q

a) min dimension is 3 of space

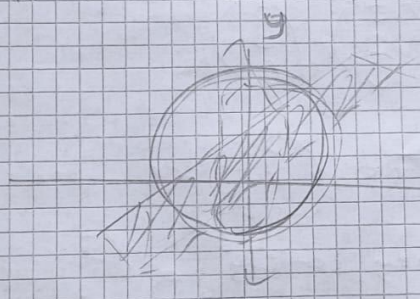
so that ∃ a chance of closed structures and volume defined only for 3 variables or more.

b) $x^2+y^2+z^2=3^2$ radius = 3 center ↓ (0,0,0)

and $x=5y$ can now betreated as a

plane = $x-5y+0z=0$

and intersection will now result in a Vol.



c) ireased geometry: $x+4y-z=6$

$x=5y$

Intersection: $5y+4y-z=6$

$z=9y-6$

let $y=k$

b)

$$\begin{bmatrix} 1 & \alpha & -\gamma & \kappa \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \kappa^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$$

These elements are zero

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 0 & 1 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

$\Rightarrow$ eliminated upper triangler matrix ✓

Q3)

Given suppose u is a unit vector in R^n

H is a $n \times n$ symmetric matrix

Such that, $H = I - 2uu^T$, As u is a unit

so $\|u\| = 1$

a) Now,

$$H^2 = H \cdot H = (I - 2uu^T)(I - 2uu^T)$$

$$= I^2 - I \cdot 2uu^T - 2uu^T \cdot I + 4uu^T \cdot uu^T$$

$$= I(I - 2uu^T - 2uu^T) + 4u \cdot \|u\|^2 \cdot u^T$$

$$= I + 2uu^T - 2uu^T + 4u \cdot \|u\|^2 \cdot u^T$$

$$\|u\| = 1$$

So $H^2 = I - 4uu^T + 4uu^T$

$$H^2 = I$$

b) $HH^T = (I - 2uu^T)(I - 2uu^T)^T$ $[(A-B)^T = A^T - B^T$

$= (I - 2uu^T)(I^T - (2uu^T)^T)$

$= (I - 2uu^T).(I - I.(u^T)^T u^T)$ $[(AB)^T = B^T.A^T$

$= (I - 2uu^T).(I - 2uu^T)$ $[As (u^T)^T = u$

$= I^2 - 2uu^T - 2uu^T + 4uu^T.uu^T$

$= I - 4uu^T + 4u\|u\|^2 u^T$ $[\|u\| = 1$

$= I - 4uu^T + 4uu^T =$

$= H.H^T = I$

Similarly

$H^T H = I$ $\quad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

so, $HH^T = H^T H = I$

so H is an orthogonal matrix

c) for a matrix A, a vector (nm-zero) will be eigen vector (suppose, v) of A corresponding eigen value λ if $Av = \lambda v$

now $Hv = (I - 2uu^T).v = Iv - 2uu^T$

$= v - 2u|u|^2$ (As I is identity matrix)

As I is a identity matrix

so, I is the only eigen value of I

and $Iu = 1.u = u$ (also) and $||u|| = +\sqrt{u^T u}$

u is a unit vector, so, $||u|| = 1 + \sqrt{u^T u}$

$u^T u = ||u||^2 = 1$

$Hu = v - 2u.1 = -v$

$\Rightarrow Hu = -v = (-1).v$

so we have $Hu = \lambda u$ where $\lambda = -1$

so, u is an eigen value for u is $= \lambda = -1$

d) given $H = (I - 2uu^T)^T$

As $H^T = (I - 2uu^T)^T$

$= I^T - 2(uu^T)^T \quad [(A-B)^T = A^T - B^T]$

$= I - 2(u^T)^T u^T$

$= I - 2(u^T)^T u^T \quad ((AB)^T = B^T \cdot A^T)$

$= I = 2uu^T = H$

So, H is a symmetric matrix and there is a theorem which stays that,

$H = Q \Lambda Q^T$ (an orthogonal matrix)

$= Q \Lambda Q^{-1}$ (As $QQ^T = I \Rightarrow Q^T = Q^{-1}$)

$= (Q^{-1})^{-1} \Lambda Q^{-1}$ (Λ = Diagonal matrix consisty of eigen value of H)

$P^{-1} \Lambda P$

so H is diagonalizable matrix.